

Probability

Definition 1.0.1 ► Discrete Probability Model

Let Ω be a set, $|\Omega| < \infty$. $\mathcal{Q} = \varphi(\Omega)$ the collection of all subsets of Ω . $P : \varphi(\Omega) \rightarrow [0, 1]$ a function. If

1. $P(\Omega) = 1$.
 2. $P(A \cup B) = P(A) + P(B)$ for all $A, B \in \varphi(\Omega)$, $A \cap B = \emptyset$.
- Then we call $(\Omega, \mathcal{Q}, P)$ a discrete probability model.

Proposition 1.0.2

Let $(\Omega, \mathcal{Q}, P)$ be a discrete probability model,

1. Let $A \in \varphi(\Omega)$, $\bar{A} := \{x \in \Omega, x \notin A\}$. Then

$$P(\bar{A}) = 1 - P(A)$$

2. $A, B \in \varphi(\Omega)$, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Remark.

If $P(\{\cdot\}) = \frac{1}{|\Omega|}$, then we say the probability is **uniform**.

Example 1.0.3

Take $\Omega = \{a, b\}$, $\mathcal{Q} = \varphi(\Omega) = \{\emptyset, \{a\}, \{b\}, \Omega\}$. $P(\{a\}) = \theta, P(\{b\}) = 1 - \theta$. Then P can extend to a probability, and $(\Omega, \mathcal{Q}, P)$ comes to a discrete probability model.

Definition 1.0.4 ► Random Variable

A **Random Variable** (R.V) is just a map $X : \Omega \rightarrow \mathbb{R}$.

Example 1.0.5 ▶ A Random Variable

$A \in \varphi(\Omega)$. Let $X = \chi_A$. Then X is a random variable. And

$$P(X = 1) = P(A), \quad P(X = 0) = 1 - P(A).$$

Definition 1.0.6 ▶ Expectation (Mean Value of a R.V)

Let X be a R.V. If

1. $|X(\Omega)| < \infty$ and $X \geq 0$. We define

$$E(X) := \sum_{x \in X(\Omega)} xP(X = x)$$

2. In general case, if $E(|X|) < \infty$, we define

$$E(X) := E(X^+) - E(X^-)$$

Proposition 1.0.7

Let $\varphi : X(\Omega) \rightarrow \mathbb{R}$. Assume that $E(|\varphi(X)|) < \infty$. Then,

$$E(\varphi(x)) = \sum_{x \in X(\Omega)} \varphi(x)P(X = x)$$

Definition 1.0.8

Suppose that $E(X^2) < \infty$. Then define the **variance** of X

$$\text{Var}(X) := E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

Proof.

$$\begin{aligned} \text{Var } X &= E[(X - E[X])^2] = E[X^2 - 2E(X)X + (E(X))^2] \\ &= E(X^2) + E(-2E(X)X) + E((E[X])^2) \end{aligned}$$

□

Definition 1.0.9 ▶ Probability Generating Function

Let X be a discrete random variable with range in $\{0, 1, 2, \dots\}$. Its Probability Generating Function, $G_X(t)$, is defined as:

$$G_X(t) := E[t^X]$$

for all real numbers t for which the expectation exists. Specifically,

$$G_X(t) = \sum_{k=0}^{\infty} t^k P(X = k)$$

Proposition 1.0.10

Assume that $R > 1$, Then

1. $E(X) = G'_X(1)$
2. $\text{Var}(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2$

Example 1.0.11 ▶ Poisson

Let $X \sim \text{Poisson}(\lambda)$,

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

$$\begin{aligned} G_X(t) &= E[t^X] = \sum_{k=0}^{\infty} t^k P(X = k) \\ &= \sum_{k=0}^{\infty} t^k \frac{\lambda^k e^{-\lambda}}{k!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(t\lambda)^k}{k!} \quad (\text{将常数和 } t \text{ 移入}) \\ G_X(t) &= e^{-\lambda} \cdot e^{t\lambda} = e^{t\lambda - \lambda} = e^{\lambda(t-1)} \\ G_X(t) &= e^{\lambda(t-1)} \end{aligned}$$